

1 Overview & System Constraints Integration

The primary objective of the Avionics and Power Subsystem is to fulfill the Powered Autonomous Delivery Aircraft (PADA) operational requirements within strict physical limits. Based on the aerodynamic constraint analysis:

- **Maximum Gross Weight ($W_{\max}$):** 19.5 oz $\approx$ 552.8 g (5.421 N).
- **Wing Area (S):** 0.08 m².
- **Baseline Wing Loading (W/S):** 67.76 N/m².

From the propulsion dynamic equations, the minimum takeoff Thrust-to-Weight ratio (T/W)_{TO} is established at 0.38. Applying a 20% safety margin for motor/propeller efficiency losses yields a design target of static $(T/W) \geq 0.50$, corresponding to a minimum static thrust requirement of $T_{\text{req}} \geq 276.4$ g.

2 Baseline Design (Initial Configuration)

Initially, a baseline avionics configuration was evaluated using standard off-the-shelf components. Table 1 summarizes the initial electrical inventory including the safety switch, active buzzer, payload camera, and airspeed sensor.

Table 1: Initial (Baseline) Electrical Components Inventory

Component	Power / Rating	Price	Weight	Selected Model / Notes
Battery	3S / 1300mAh (14.43Wh)	1,000 EGP	115 g	LiPo 3S 1300mAh 45C (< 100 Wh)
ESC	30A Cont. (40A Burst)	600 EGP	28 g	30A Brushless ESC with 5V/3A BEC
Motor	180W / Max 18A	900 EGP	55 g	Sunnysky X2212 KV1400
Flight Controller	5V / 500mA	2,250 EGP	38 g	Pixhawk 4 Mini / Matek H743
Receiver	5V / 100mA	750 EGP	8 g	900MHz Receiver (ExpressLRS)
GPS	5V / 150mA	900 EGP	22 g	Holybro Micro M8N GPS Module
Telemetry	500mW / 5V	1,100 EGP	18 g	Telemetry Transceiver
Servos (x5)	1.8 kg-cm Torque	750 EGP	45 g	SG90 Micro Servos (1–2 kg · cm)
Airspeed Sensor	5V / 25mA	850 EGP	10 g	Digital Pitot Tube Sensor Kit
FPV / Payload Cam	5V / 200mA	650 EGP	12 g	Micro FPV / Delivery Camera
Buzzer & E-Switch	5V / Active Piezo	150 EGP	8 g	Active Buzzer & Safety Arming Switch
Total	–	9,900 EGP	359 g	Leaves 193.8g for Airframe

3 Optimization & Trade-Off Analysis

While the initial configuration met all electrical and thrust demands, its cumulative mass of **359 g** consumed too much of the maximum gross weight limit (552.8 g). To resolve this, an avionics optimization trade study was conducted:

1. **Battery Downsizing (1300 mAh $\rightarrow$ 850 mAh):** Saving **40 g** while staying strictly under the < 100 Wh rule.

2. **Motor Sizing Adjustment** (2212 $\rightarrow$ 2204): Saving **27 g** while preserving $(T/W) \approx 0.76$.
3. **ESC Optimization** (30A $\rightarrow$ 20A): Saving **16 g** with a 42% safety margin.
4. **Flight Controller Integration** (Matek H743-WING): Saving **8 g** and integrating internal power distribution and current sensing.

4 Optimized Final Design (Current Configuration)

Table 2 presents the final electrical component inventory following the trade study reductions.

Table 2: Optimized (Final) Electrical Components Inventory

Component	Power / Rating	Price	Weight	Selected Model / Rationale
Battery	3S/850mAh (9.44Wh)	800 EGP	75 g	LiPo 3S 850mAh 45C (< 100 Wh rule)
ESC	20A Cont. / 25A Burst	450 EGP	12 g	20A BLHeli_S ESC with 5V BEC
Motor	KV2300 / 14A	650 EGP	28 g	2204 Outrunner (Yields $T/W = 0.76$)
Flight Controller	5V / 500mA	2,250 EGP	30 g	Matek H743-Wing (ArduPilot/MAVROS)
Receiver	5V / 100mA	750 EGP	8 g	900MHz Receiver (JST-GH)
GPS	5V / 150mA	900 EGP	20 g	Holybro Micro M8N GPS
Telemetry Radio	500mW / 5V	1,100 EGP	18 g	Telemetry Transceiver
Servos (x5)	1.8 kg-cm Torque	750 EGP	45 g	SG90 Micro Servos (1–2 kg · cm)
Airspeed Sensor	5V / 25mA	850 EGP	10 g	MS4525DO Digital Pitot Sensor
FPV / Payload Cam	5V / 200mA	650 EGP	12 g	Micro FPV / Delivery Camera
Buzzer & E-Switch	5V / Active Piezo	150 EGP	8 g	Active Buzzer & Safety Arming Switch
Total	–	9,300 EGP	266 g	Leaves 286.8g for Airframe

5 Detailed Power Budget and Flight Endurance Calculations

To mathematically prove the electrical feasibility and endurance of the optimized PADA system, the power and energy equations are structured as follows:

5.1 1. Total Battery Energy Capacity (E_{batt})

Using the nominal voltage ($V_{\text{nom}} = 11.1 \text{ V}$) and capacity ($C = 0.85 \text{ Ah}$):

$$E_{\text{batt}} = V_{\text{nom}} \times C = 11.1 \text{ V} \times 0.85 \text{ Ah} = 9.435 \text{ Wh}$$

This satisfies the project constraint of being strictly below 100 Wh and avoids high-voltage LiHV chemistries.

5.2 2. Maximum Continuous Current and Discharge Safety Check

The battery discharge rate is 45C with an 850 mAh (0.85 A) rating:

$$I_{\text{max, discharge}} = C_{\text{rate}} \times \text{Capacity} = 45 \times 0.85 \text{ A} = 38.25 \text{ A}$$

Since the maximum system current draw under full throttle does not exceed 14 A (motor peak 14 A + avionics baseline ~ 0.8 A), the battery operates well within its safe discharge threshold ($14 \text{ A} < 38.25 \text{ A}$).

5.3 3. Average System Current Draw ($I_{\text{total, avg}}$)

During normal cruise conditions, the motor draws an average of 5.5 A, while the avionics suite draws an aggregate baseline current of ≈ 0.8 A:

$$I_{\text{total, avg}} = I_{\text{motor, cruise}} + I_{\text{avionics, baseline}} = 5.5 \text{ A} + 0.8 \text{ A} = 6.3 \text{ A}$$

5.4 4. Estimated Flight Endurance (T_{flight})

Applying an 80% Depth of Discharge (DOD) factor to preserve battery longevity:

$$T_{\text{flight}} = \left(\frac{\text{Capacity} \times \text{DOD}}{I_{\text{total, avg}}} \right) \times 60 = \left(\frac{0.85 \text{ Ah} \times 0.80}{6.3 \text{ A}} \right) \times 60 \approx 6.47 \text{ minutes}$$

This provides a sufficient operational window for autonomous mission execution, waypoint navigation, obstacle avoidance, and payload deployment.

6 Wiring Topology, Exact Port Mapping, and Connectors

Table 3 outlines the precise physical connections, wire types, gauges (AWG), and Matek H743-WING port mapping matching the schematics architecture cleanly within the page limits.

Table 3: Matek H743-WING Port Mapping and Wiring Specifications

Subsystem Device	Controller Port	Wire Gauge	Type	&	Connection Notes / Safety
Battery to PDB/ESC	Battery Pads (+BAT/ - BAT)	14 AWG Wire	Silicone		Main power input with XT60 & Cap (470 μ F/35V)
Brushless Motor	ESC Motor Outputs (U, V, W)	16 AWG Wire	Silicone		Secure bullet connectors soldered to ESC
ESC Signal	ESC / S1 PWM Pad + GND + 5V	26/24 AWG / Silicone	PWM		Ground, 5V BEC, and S1 PWM signal
GPS Module	UART2 (TX2/RX2/5V/GND)	28 AWG 4-pin	JST-GH		Connected via JST-GH 4Pins to UART2
Compass (Mag)	I2C1 (CL1/DA1/5V/GND)	28 AWG 4-pin	JST-GH		Connected via JST-GH 4Pins to I2C1 bus
900MHz Receiver	UART1 (TX1/RX1/5V/GND)	28 AWG 4-pin	JST-GH		Connected via JST-GH 4Pins to UART1
Telemetry Radio	UART7 (TX7/RX7/5V/GND)	28 AWG 4-pin	JST-GH		Connected via JST-GH 4Pins to UART7
Servo 1 (Left Aileron)	PWM Output Pad S2	24 AWG Wire	Silicone		Direct PWM control signal
Servo 2 (Right Aileron)	PWM Output Pad S3	24 AWG Wire	Silicone		Direct PWM control signal
Servo 3 (Elevator)	PWM Output Pad S4	24 AWG Wire	Silicone		Direct PWM control signal
Servo 4 (Rudder)	PWM Output Pad S5	24 AWG Wire	Silicone		Direct PWM control signal
Servo 5 (Flaps)	PWM Output Pad S6	24 AWG Wire	Silicone		Direct PWM control signal
Payload Camera	Video IN (VIN) + 5V/GND	28 AWG Coaxial	Shielded		Real-time FPV feed and visual target
Buzzer & E-Switch	BUZ, GND, LED, Safety Pads	28 AWG	JST Connectors		Mandatory safety switch and buzzer

7 Summary of Optimization Impact

Through this engineering optimization, the total avionics mass was reduced from **359 g to 266 g** (a **25.9% mass reduction**), saving **93 g** in total. This increases the allowable airframe structural mass allocation to **286.8 g**, ensuring sufficient structural integrity without exceeding the 552.8 g maximum gross weight limit.

8 Master Avionics Wiring Schematic

Figure 1 displays the integrated electrical pinout, power bus routing, and sensor bus topology for the PADA UAV system using the Matek H743-WING controller matching the exact routing layout.

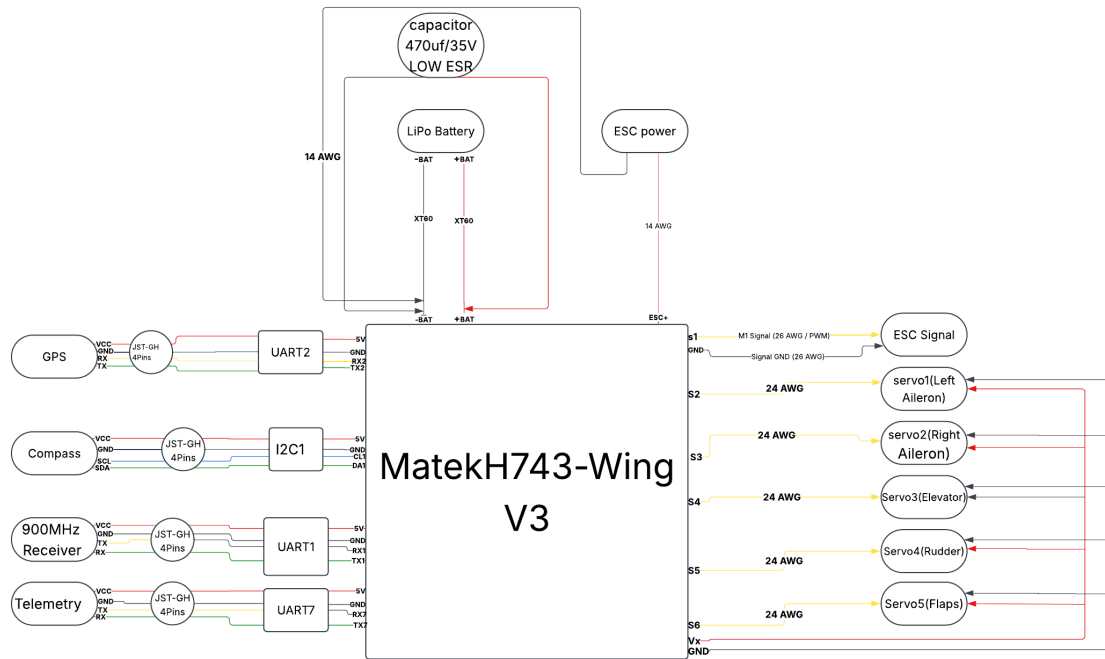


Figure 1: PADA Master Avionics and Power Schematic Mapping.